

Minor Project

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Introduction

- Automotive Safety Crisis: Driver fatigue causes over 20% of commercial vehicle crashes worldwide, making real-time Driver Monitoring Systems (DMS) essential.
- Facial Behavioral Analysis: Uses non-invasive computer vision to analyze eye openness (EAR), yawning (MAR), PERCLOS, and head pose angles.
- The Non-IoT Edge Approach: Executes 100% on-device AI inference directly on embedded hardware (Raspberry Pi 5 / NVIDIA Jetson Orin Nano).
- Key Technical Advantages: Delivers sub-5ms latency, complete offline reliability, zero cellular bandwidth cost (\$0/mo), and total data privacy (GDPR compliance).

Background and Motivation

- Flaws of Cloud/IoT Architectures: Cloud-dependent systems suffer from network latency jitter (150ms–500ms+), rendering micro-sleep alerts ineffective.
- Connectivity Dead Zones: Vehicles frequently navigate tunnels, mountain passes, and rural highways where 4G/5G cellular signal is lost.
- Biometric Privacy Liabilities: Transmitting high-definition facial video streams to cloud servers poses severe GDPR and biometric privacy risks.
- High Operating Expense: Cloud video streaming costs \$15–\$30/month per vehicle in cellular data and cloud server compute fees.
- Edge Motivation: An autonomous, sub-5ms on-device system guarantees 100% alert reliability anywhere without recurring costs.

Relevance and Importance

- Sub-5ms Compression Pipeline: Combines MediaPipe BlazeFace + Face Mesh backbones with Quantization-Aware Training (QAT INT8) for sub-5ms inference.
- 24/7 NIR Lighting Robustness: Implements 850nm / 940nm Near-Infrared illumination with adaptive CLAHE contrast enhancement for day & night operation.
- Eyewear & Sunglasses Robustness: Features dual-wavelength NIR penetration and dynamic fallback to Mouth Aspect Ratio (MAR) + Head Nodding when eyes are occluded.
- Personalized Calibration (<10s): Automatically calculates baseline thresholds ($\text{EAR_thresh} = \text{mean} - 2.5 * \text{std}$) without manual configuration.
- Commercial Relevance: Reduces fleet accident rates, lowers insurance liabilities, and scales easily across commercial trucking, buses, and ride-shares.

Literature Review

- Soukupova & Cech (2016) [R1]: Introduced Eye Aspect Ratio (EAR) using 68 facial landmarks for real-time blink and eye closure detection.
- Dinges & Grace (1998) [R2]: Established PERCLOS (Percentage of Eye Closure $\geq 80\%$ over time window) as the gold standard fatigue metric.
- Garces et al. (2021) [R5]: Formulated Mouth Aspect Ratio (MAR) to quantify yawning magnitude and duration.
- Dzyakanchuk et al. (2022) [R3]: Evaluated low-power CNN face detection on Raspberry Pi embedded hardware.
- Research Gap Addressed: Existing literature lacks sub-5ms INT8 quantized multi-metric landmark pipelines with NIR robustness on sub-\$100 edge hardware.

Objectives

- Develop a 100% offline Non-IoT Edge driver monitoring system executing in under 5 ms frame latency (200+ FPS capability).
- Implement mathematical models for Eye Aspect Ratio (EAR), Mouth Aspect Ratio (MAR), PERCLOS, and 3D Head Pose Euler angles.
- Design a 4-stage compression pipeline (Architecture Selection -> Knowledge Distillation -> Structured Pruning -> QAT INT8).
- Achieve 24/7 lighting robustness using active NIR (850nm/940nm) illumination and adaptive CLAHE contrast enhancement.
- Provide eyewear robustness with an automated fallback scoring matrix for dark and mirrored sunglasses.
- Deploy and benchmark on sub-\$100 hardware (RPi 5 + Google Coral / Hailo-8L) and NVIDIA Jetson Orin Nano.

Required Tools and Technology

- Hardware Platforms: Raspberry Pi 5 (\$35), Google Coral USB Edge TPU (\$60), Hailo-8L 13 TOPS HAT (\$70), NVIDIA Jetson Orin Nano (\$249).
- Optical Hardware: Near-Infrared (NIR) 850nm/940nm LED arrays, narrow bandpass optical filters, No-IR-Cut camera sensor.
- Programming Runtimes: Python 3.10+ & C++17 for high-performance multithreaded frame ingestion and inference.
- Vision & AI Runtimes: OpenCV 4.8+ (CLAHE, V4L2 pipeline), MediaPipe Tasks API (468 3D landmarks), TFLite INT8 (XNNPACK), TensorRT 8.6+, HailoRT.
- ML Classifiers & HMI: Scikit-Learn (SVM RBF classifier for 4-level fatigue grading), RPi.GPIO / python-can for physical buzzer & CAN bus alert output.

Method/ Approach

- Stage 1 - Preprocessing: V4L2 NIR frame capture (1080p@30fps) -> Color space conversion -> Adaptive CLAHE (clipLimit=4.0, grid 8x8).
- Stage 2 - Edge AI Inference: BlazeFace face detection (1.5ms) -> 468-point Face Mesh INT8 landmark extraction (2.5ms).
- Stage 3 - Metric & Calibration: Compute EAR, MAR, PERCLOS, and 3D Head Pose via cv2.solvePnP(). Auto-calibrate baseline in < 10 seconds.
- Stage 4 - Compression Pipeline: Teacher ResNet-50 distillation -> 30% structured channel pruning -> 5-epoch Quantization-Aware Training (QAT INT8).
- Stage 5 - Classification & Alerting: 3-Feature SVM (PERCLOS + EAR + MAR) evaluates fatigue severity (Level 0: Alert -> Level 3: Micro-sleep Hazard) in < 0.1ms.

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